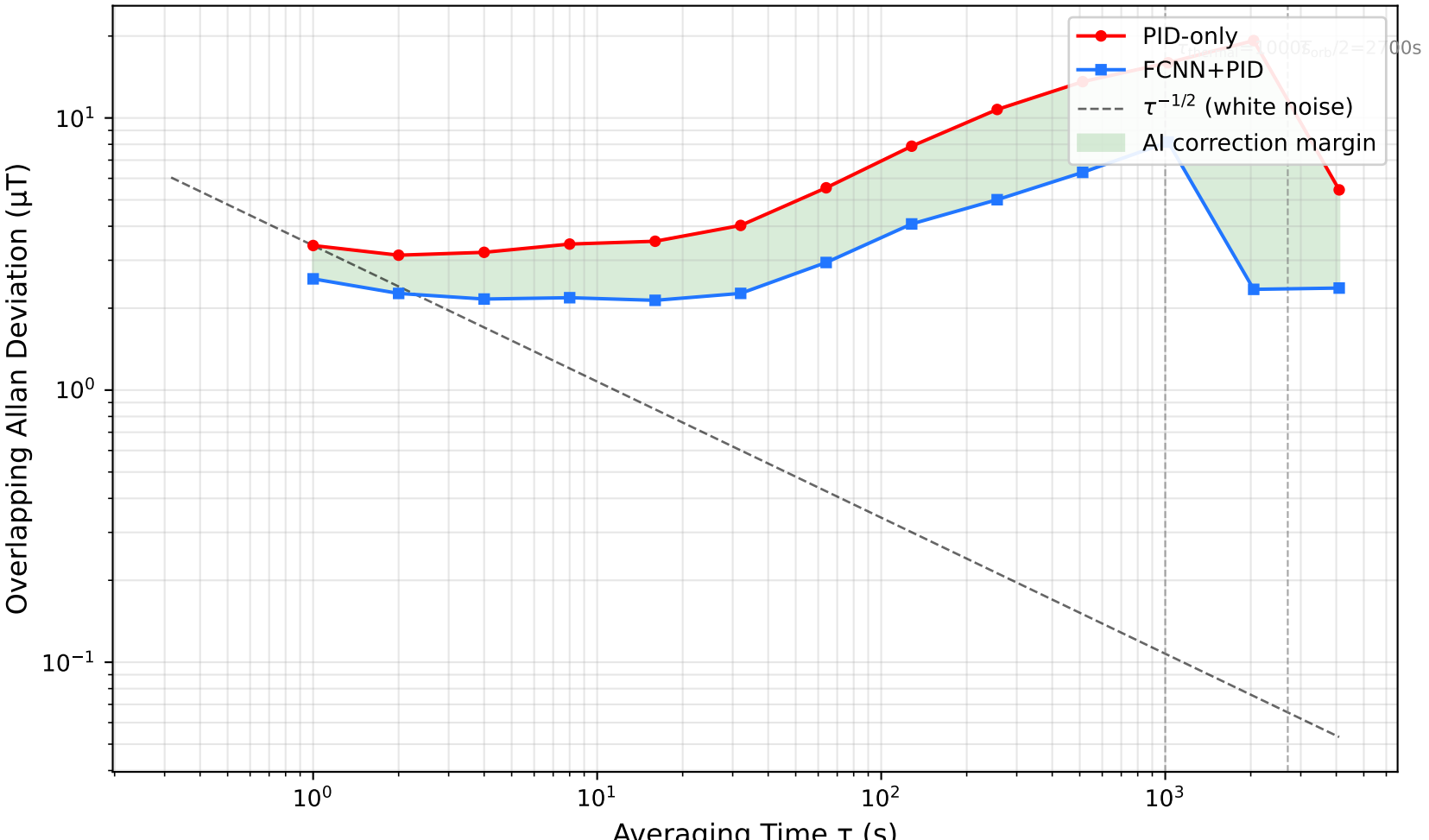
Measurement Stability: PID-only vs FCNN+PID Hybrid
(3 LEO orbits, 50 μT, satellite tumble)
PID-only
FCNN+PID
$\tau^{-1/2}$ (white noise)
AI correction margin
Overlapping Allan Deviation (μT)
Averaging Time τ (s)
10^{1}
10^{0}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}